Implementation of Multivariate Linear Regression

Aim

To write a python program to implement multivariate linear regression and predict the output.

Equipment's required:

1. Hardware – PCs
2. Anaconda – Python 3.7 Installation / Moodle-Code Runner

Algorithm:

Step1 import pandas as pd.

Step2 Read the csv file.

Step3 Get the value of X and y variables

Step4 Create the linear regression model and fit.

Step5 Predict the CO2 emission of a car where the weight is 2300kg, and the volume is 1300cm cube.

Program:

```
Developed by : AMIRDAVARSHINI D
Reg No : 212225230013
import pandas as pd
```

```
from sklearn import linear_model
df = pd.read_csv("car.csv")
X = df[['Weight', 'Volume']]
y = df['CO2']
regr = linear_model.LinearRegression()
regr.fit(X, y)
print('Coefficients:', regr.coef_)
print('Intercept:', regr.intercept_)
predictedCO2 = regr.predict(pd.DataFrame([[3300, 1300]], columns=['Weight', 'Volume']))
print('Predicted CO2 for the corresponding weight and volume:', predictedCO2)
```

Output:

Insert your output

```
Coefficients: [0.00755095 0.00780526]
Intercept: 79.69471929115939
Predicted CO2 for the corresponding weight and volume: [114.75968007]
```

Result

Thus the multivariate linear regression is implemented and predicted the output using python program.